Edge-First Triage & Segmentation for Gigapixel Pathology

VAE-Triage (WSI QC + GMM tissue components) · High-resolution U-Net segmentation

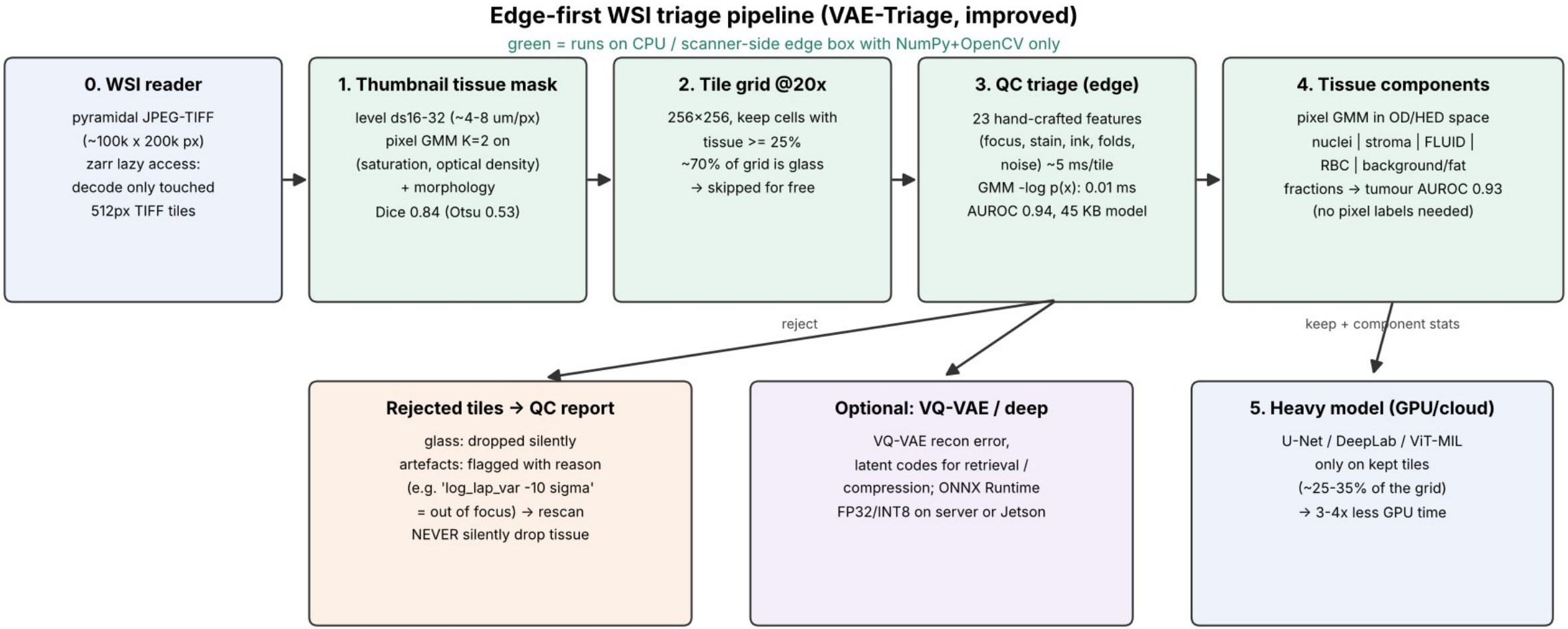
- Real data: 7 CAMELYON16 breast lymph-node WSIs (203k tiles @20x) + ISPRS Potsdam
- Everything measured on a 4-core CPU – no GPU – i.e. edge conditions
- GMM-based, label-free, explainable gate in front of heavy deep models

Problem

A WSI is ~10^10 pixels; most of it should never reach the GPU

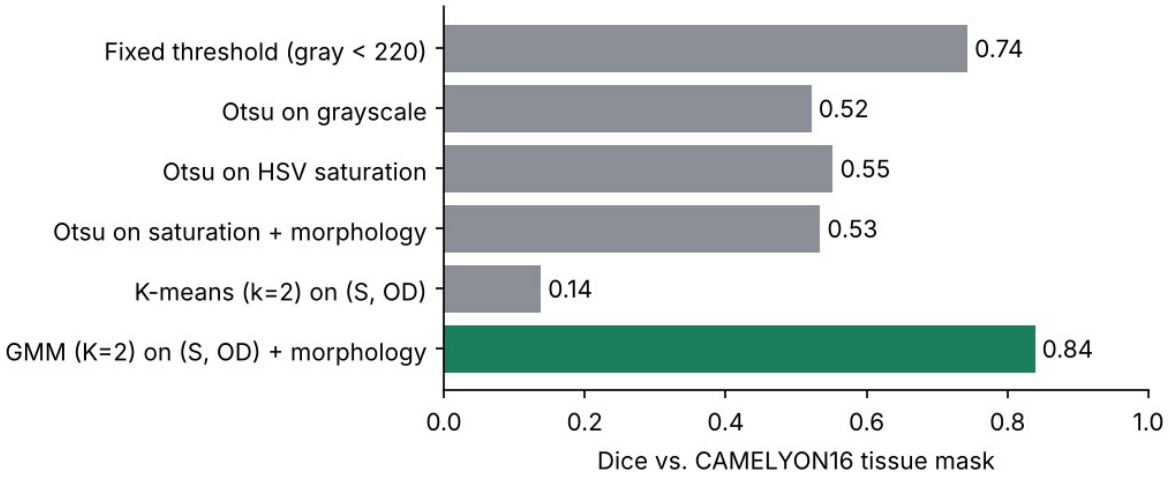
- Pyramidal JPEG-TIFF, 100k × 200k px, 1–4 GB per slide
- 69.6% of 203,168 tiles in our 7 slides are empty glass (30–78% per slide)
- Some tissue tiles are unusable: blur, pen ink, folds, bubbles, exposure, JPEG
- Heavy models (U-Net / DeepLab / ViT-MIL) cost 10–100 ms per tile on GPU
- Goal: decide cheaply, per tile, at the edge: tissue? usable? what's in it?

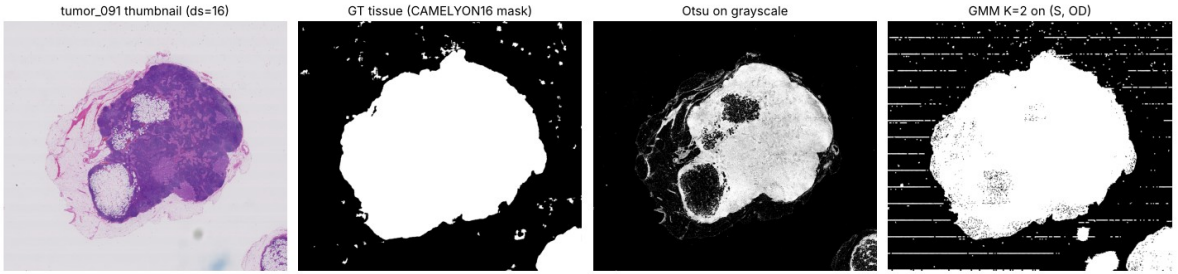
Pipeline


Edge-first WSI triage pipeline (VAE-Triage, improved)
green = runs on CPU / scanner-side edge box with NumPy+OpenCV only
0. WSI reader
pyramidal JPEG-TIFF
(~100k x 200k px)
zarr lazy access:
decode only touched
512px TIFF tiles
1. Thumbnail tissue mask
level ds16-32 (~4-8 um/px)
pixel GMM K=2 on
(saturation, optical density)
+ morphology
Dice 0.84 (Otsu 0.53)
2. Tile grid @20x
256×256, keep cells with
tissue >= 25%
~70% of grid is glass
→ skipped for free
3. QC triage (edge)
23 hand-crafted features
(focus, stain, ink, folds,
noise) ~5 ms/tile
GMM -log p(x): 0.01 ms
AUROC 0.94, 45 KB model
4. Tissue components
pixel GMM in OD/HED space
nuclei | stroma | FLUID |
RBC | background/fat
fractions → tumour AUROC 0.93
(no pixel labels needed)
reject
keep + component stats
Rejected tiles → QC report
glass: dropped silently
artefacts: flagged with reason
(e.g. 'log_lap_var -10 sigma'
= out of focus) → rescan
NEVER silently drop tissue
Optional: VQ-VAE / deep
VQ-VAE recon error,
latent codes for retrieval /
compression; ONNX Runtime
FP32/INT8 on server or Jetson
5. Heavy model (GPU/cloud)
U-Net / DeepLab / ViT-MIL
only on kept tiles
(~25-35% of the grid)
→ 3-4x less GPU time

Stage 0 – tissue detection on the thumbnail

Pixel GMM (K=2) on saturation + optical density

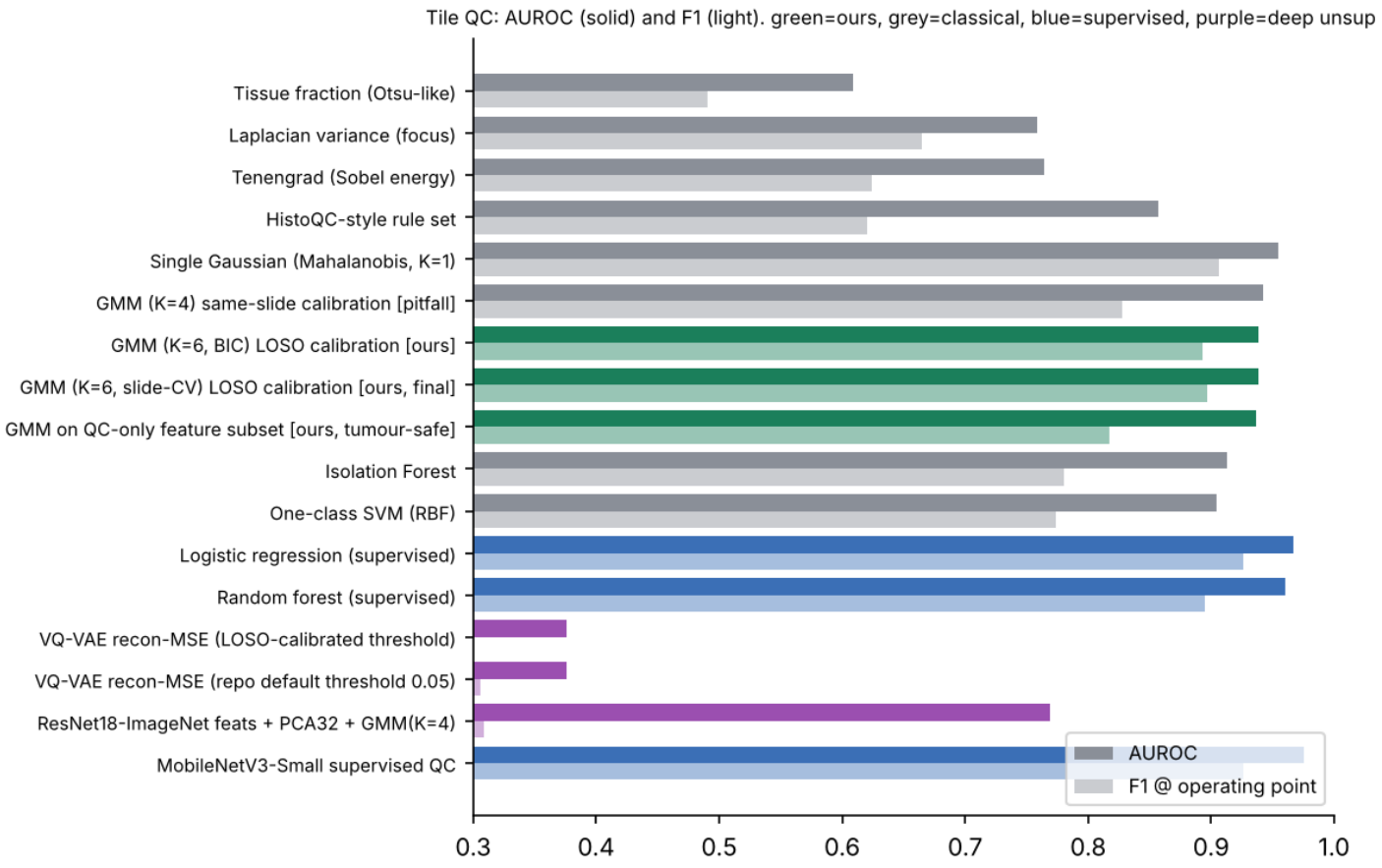




- GMM Dice 0.84 vs Otsu 0.53 vs k-means 0.14
- Glass = tight cluster, tissue = broad cluster → needs unequal covariances

Stage 1 – tile QC with a GMM density

23 physics features → −log p(x), threshold = 95% clean pass on UNSEEN slides



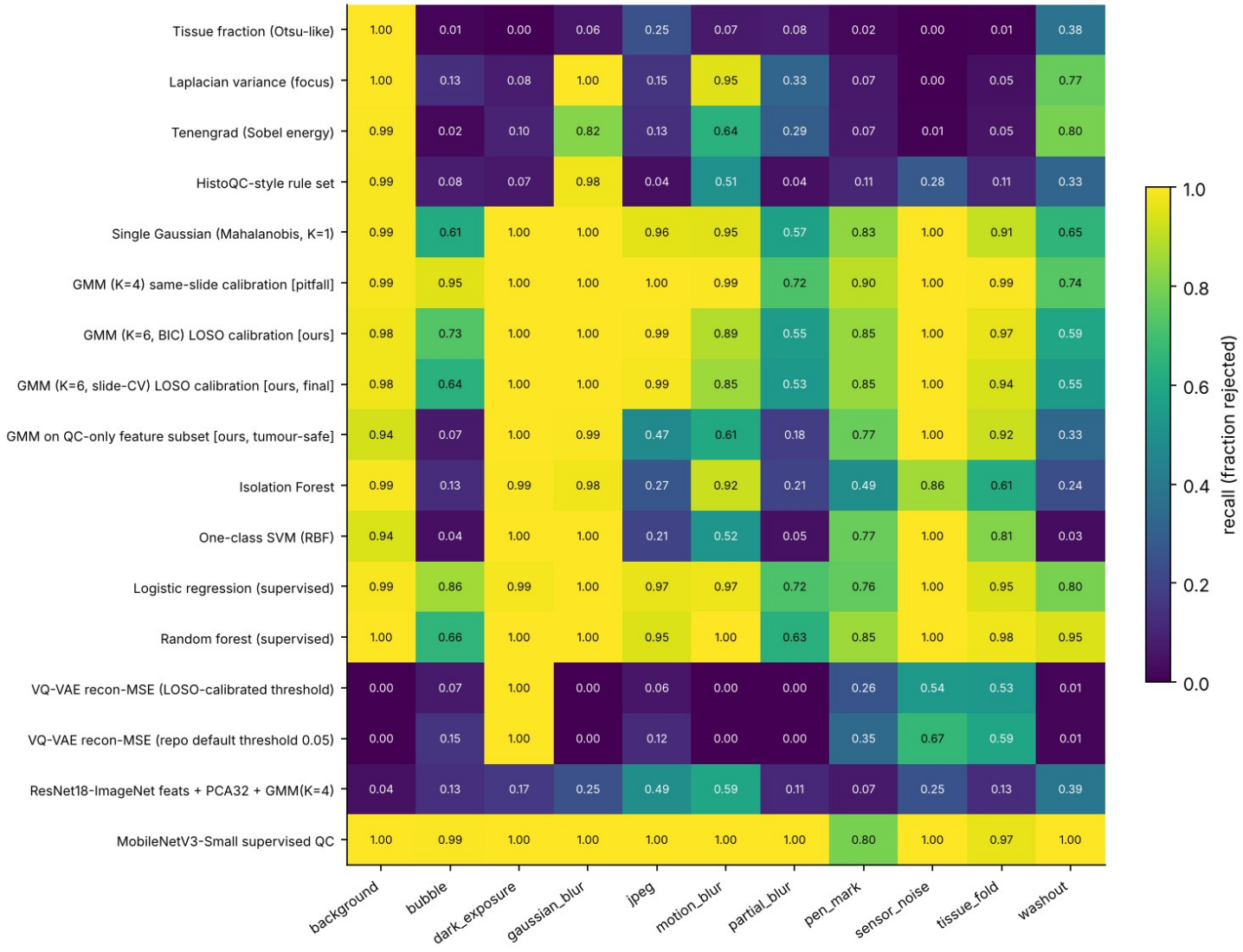
- GMM: AUROC 0.939, F1 0.897, spec 0.928
- Single Gaussian 0.954 (few slides → simplest density wins)
- Laplacian 0.76 · HistoQC-style 0.86 · IsoForest 0.91
- Supervised MobileNetV3 0.976 (needs labels)
- 45 KB model · 0.01 ms score · 2.1 ms features (128 px)

Why unsupervised: artefacts never seen in training

Model	bubble	pen ink	fold	JPEG
Logistic regression (sup.)	0.13	0.64	0.41	0.46
Random forest (sup.)	0.07	0.64	0.24	0.22
GMM K=1 (unsup.)	0.61	0.83	0.91	0.96
GMM K=6 (unsup.)	0.64	0.85	0.94	0.99

- Production scanners create failure modes nobody annotated → novelty detection
- Best design: GMM gate for the unknown + small supervised head for known artefacts

Per-artefact recall

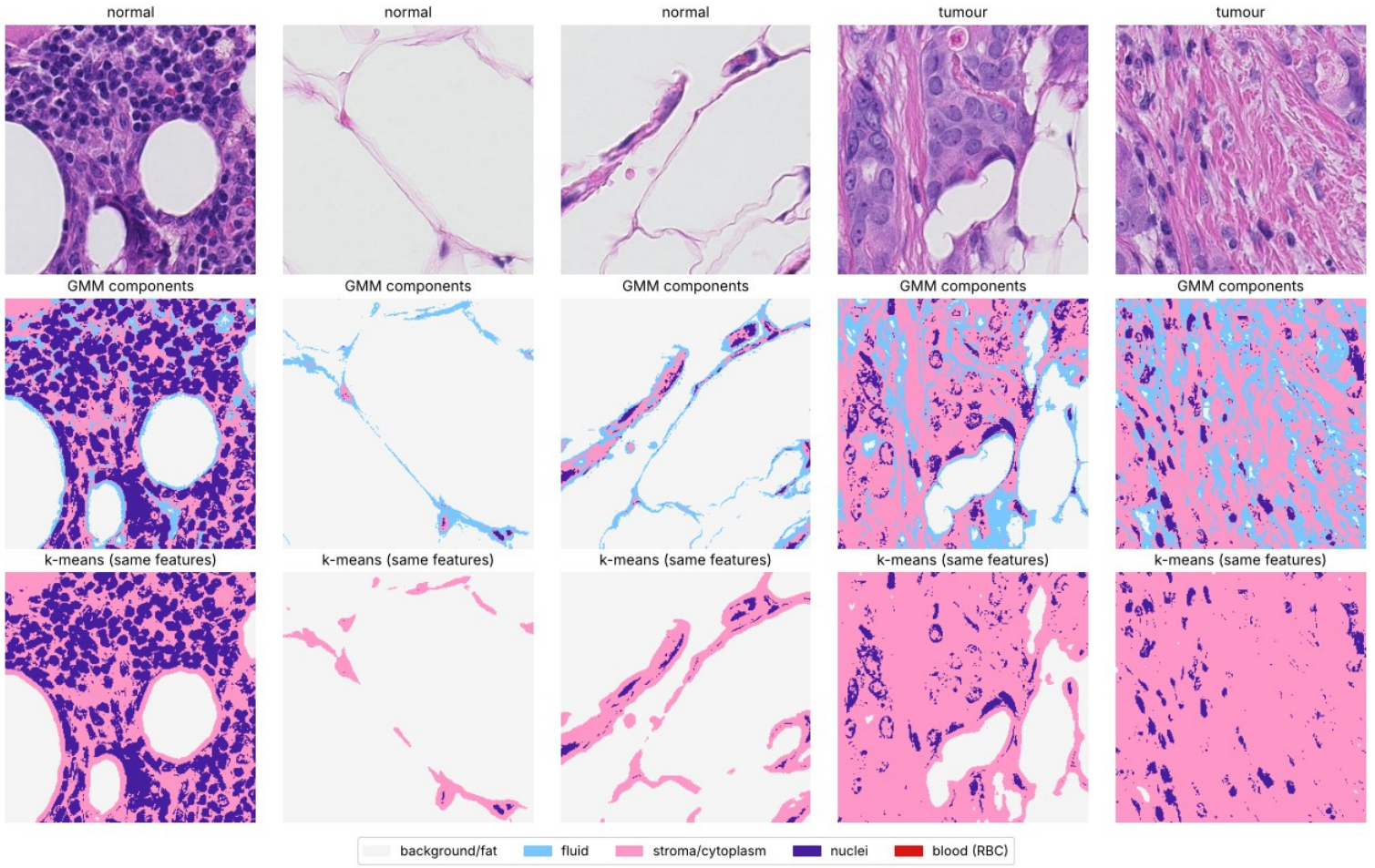


Two traps we found (and fixed)

- Calibration trap: threshold from tiles of the SAME slides
  - specificity 0.69 and 43% of tumour tiles rejected on new slides
  - leave-one-slide-out calibration → specificity 0.93, tumour rejected 3.5%
- VQ-VAE reconstruction error as QC score: AUROC 0.376 (worse than chance)
  - blurred / empty tiles are smooth → EASIER to reconstruct → look 'normal'
  - keep VQ-VAE as compressor/tokeniser + tumour-novelty signal (0.749)
- BIC over-estimates K: tiles of one slide are not i.i.d. → select K on held-out slides

Stage 2 – GMM tissue components incl. FLUID

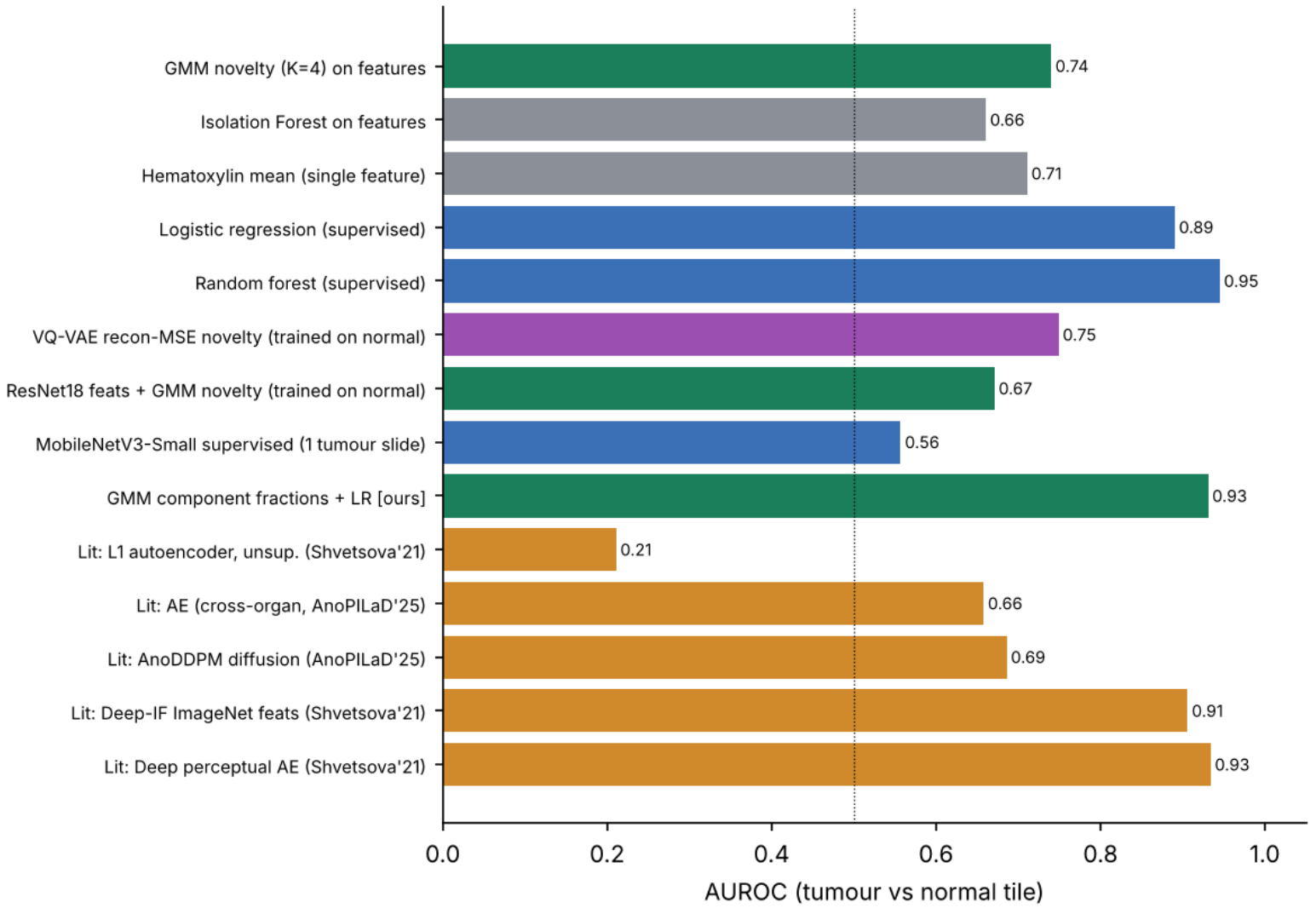
Beer–Lambert → OD → colour deconvolution (H, E) → pixel GMM



- Components: nuclei · stroma/cytoplasm · fluid · blood · fat/glass
- Fluid = pale, eosin-dominant, low OD (plasma, oedema, proteinaceous fluid)
- k-means on same features: no fluid cluster
- Cross-slide stability: 79% pixel agreement
- 5 fractions → tumour AUROC 0.93

Tumour vs normal tiles

Unsupervised scores vs supervised vs literature



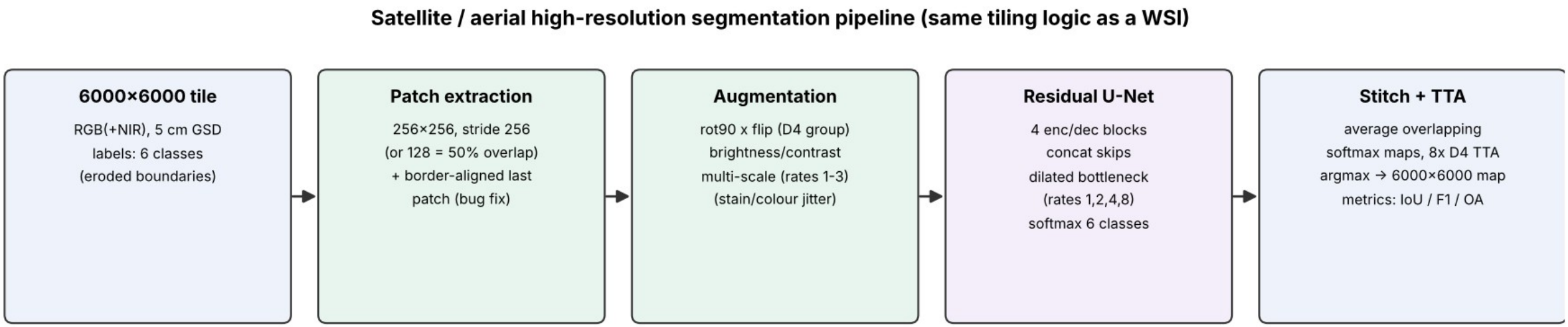
Edge cost (measured, 4-vCPU, no GPU)

Component	ms / tile	Size	Note
Features (23-D) + GMM	2.1 + 0.01 ms	45 KB	no labels, explainable
VQ-VAE ONNX Runtime FP32 / INT8	22.9 / 22.7 ms	2.9 / 0.8 MB	INT8: smaller, not faster
MobileNetV3-S ORT FP32 / INT8	0.6 / 1.4 ms	6.1 / 1.7 MB	needs labels per artefact
ResNet-18 features (PyTorch)	15.8 ms	45 MB	generic, weak for QC

- Slide with 29k grid tiles → ~8.8k tissue tiles → QC in ~5 s on 4 cores
- Only ~30% of the grid reaches the GPU model → 3–4× less compute
- Always measure on target hardware (INT8 was slower on this CPU)

Project 2 – high-resolution segmentation

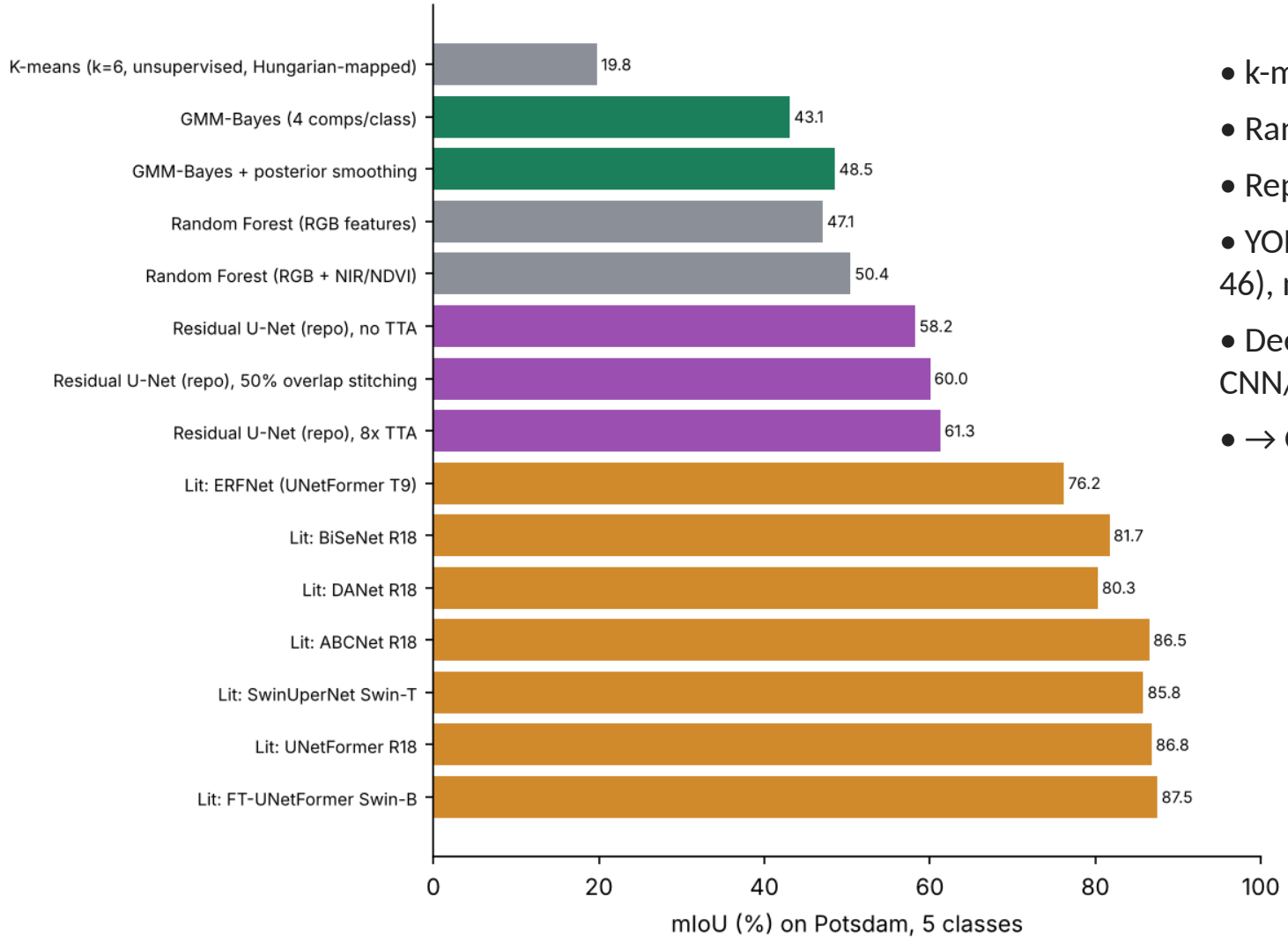
Residual U-Net + dilated bottleneck, patch stitching, 8× TTA



- Same problem as WSI segmentation: image ≫ network input
- Bugs fixed: last 112 px of each 6000-px tile never predicted (3.7% of pixels); num_classes fell back to 5
- Terminology: 'residual' blocks are concatenations (DenseNet-style)

Potsdam results vs literature

Classical pixel classifiers (minutes on CPU) vs published deep models



- k-means 19.8 → GMM-Bayes 43.1 → + smoothing 48.5
- Random forest 47.1 → + NIR/NDVI 50.4
- Repo U-Net, 30 min CPU: 58.2 → 61.3 with 8× TTA
- YOLOv8n-seg cars (20 min CPU): car IoU 84.7 (U-Net 61, RF 46), mask mAP50 0.85
- Deep SOTA 86.5–87.5: shape & context need a CNN/transformer
- → GMMs for colour/density problems, U-Net for semantics

What I changed and what's next

- Done: fixed crashes, real WSI reader, tissue mask, GMM QC + export, LOSO calibration,
  - GMM tissue components (fluid), explainable rejections, benchmarks on real slides
- Next: stain augmentation + many slides for supervised heads (1-slide CNN → 0.56 AUROC)
- Next: feature-space (perceptual) anomaly score instead of pixel MSE (lit. 0.66 → 0.93)
- Next: U-Net with pretrained encoder, 50% overlap + Gaussian blending, Dice+CE loss
- Next: export GMM to C/ONNX for scanner software; GMM likelihood as drift monitor

60-second summary

- 70% of a slide is glass: a K=2 GMM on the thumbnail removes it (Dice 0.84)
- A GMM density on 23 physics features flags bad tiles: AUROC 0.94, 45 KB, explainable
- It catches artefact types it never saw (64–99% vs 7–64% for supervised)
- A pixel GMM in optical-density space separates nuclei, stroma, fluid, blood, fat
- Only good tissue reaches the U-Net – cheap decisions at the edge, expensive ones on the GPU